

Positive Factorization Through Free Banach Lattices For Atomic ℓ_1 Retracts

Automatic math-discovery smoke test

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Source Problem

The source paper is Timur Oikhberg, *Geometry of unit balls of free Banach lattices, and its applications*, arXiv:2303.05209. The relevant discussion appears in Section 3, on pages 12–13 of the arXiv PDF. The paper defines positive factorization through a free Banach lattice and then asks whether the canonical factorization

$$X \longrightarrow \text{FBL}[X] \longrightarrow X$$

for an arbitrary Banach lattice can always be made with a positive first map.

$p < \infty$, then $\text{FBL}^{(p)}[E]$ has λK_p -Positive Approximation Property.

It is an open problem whether any Banach lattice with the λ -AP must also have ν -PAP, for some ν . To partially resolve this, we introduce a new definition: we say that a Banach lattice X *λ -positively factors* through $\text{FBL}^{(p)}[X]$ if there exist positive maps $u : X \rightarrow \text{FBL}^{(p)}[X]$ and $v : \text{FBL}^{(p)}[X] \rightarrow X$ so that $I_X = vu$, and $\|u\|\|v\| \leq \lambda$. λ -positive factorization through $\text{FBL}^{\uparrow p}[E]$ is defined similarly.

Theorem 3.4 immediately implies:

In [3], a stronger property was considered: a Banach lattice X is said to have the *λ -lattice lifting property* (through $\text{FBL}^{(p)}[X]$ or $\text{FBL}^{\uparrow p}[X]$) if, in the definition of positive factorization, u and v can be taken to be lattice homomorphisms. By [3, Proposition 2.3], for any Banach space E , $\text{FBL}^{(p)}[E]$ ($\text{FBL}^{\uparrow p}[E]$) has the 1-lattice lifting through $\text{FBL}^{(p)}[\text{FBL}^{(p)}[E]]$ (respectively, $\text{FBL}^{\uparrow p}[\text{FBL}^{\uparrow p}[E]]$) (the proof covers $\text{FBL}[E]$, but generalizes easily). This implies:

Corollary 3.7. *For a Banach space E , and $1 \leq p \leq \infty$, $\text{FBL}^{(p)}[E]$ has the μ -Approximation Property if and only if it has the μ -Positive Approximation Property.*

In general, for every Banach lattice X , there exist a contraction $u : X \rightarrow \text{FBL}[X]$, and a contractive lattice homomorphism $v : \text{FBL}[X] \rightarrow X$, so that $I_X = vu$. We do not know whether we can always achieve a factorization with a positive u . By [21], certain lattices X do not lattice embed into any free lattice, so we cannot take u to be a lattice homomorphism in the above factorization (no matter what v is).

Result Type

This packet gives a **partial result**. It proves the desired positive factorization for Banach lattices that are positive retracts of an atomic $\ell_1(I)$. It does not settle the arbitrary Banach-lattice question.

Proof Intuition

For $\ell_1(I)$, the positive coordinates give an explicit lifting into the free Banach lattice: send the i -th unit vector to the positive part of the i -th free generator. This is linear on $\ell_1(I)$, positive, and contractive because the series is absolutely summable in the free-lattice norm. The canonical lattice homomorphism back to $\ell_1(I)$ sends that positive part to the original positive atom. If X is a positive retract of $\ell_1(I)$, functoriality of the free Banach lattice transfers this lifting from $\ell_1(I)$ to X .

Definitions

For a Banach space E , write $\iota_E : E \rightarrow \text{FBL}[E]$ for the canonical isometric embedding. If X is a Banach lattice, let

$$\rho_X : \text{FBL}[X] \longrightarrow X$$

be the contractive lattice homomorphism extending the identity map on X . Thus $\rho_X \iota_X = I_X$. If $T : E \rightarrow F$ is a bounded operator between Banach spaces, $\text{FBL}[T] : \text{FBL}[E] \rightarrow \text{FBL}[F]$ denotes the induced lattice homomorphism satisfying

$$\text{FBL}[T] \iota_E(e) = \iota_F(Te).$$

This is the standard universal property of free Banach lattices [1, 2].

A Banach lattice X is called a positive retract of $\ell_1(I)$ here if there are positive bounded linear maps

$$J : X \rightarrow \ell_1(I), \quad Q : \ell_1(I) \rightarrow X, \quad QJ = I_X.$$

Main Result

Lemma 1. *For every index set I , $\ell_1(I)$ has a 1-positive factorization through $\text{FBL}[\ell_1(I)]$.*

Proof. Let $(e_i)_{i \in I}$ be the canonical positive unit vectors of $\ell_1(I)$, and let

$$\rho : \text{FBL}[\ell_1(I)] \rightarrow \ell_1(I)$$

be the lattice homomorphism extending the identity map on $\ell_1(I)$. For $a = (a_i) \in \ell_1(I)$, define

$$Sa = \sum_{i \in I} a_i \iota_{\ell_1(I)}(e_i)^+.$$

The sum is the norm limit of its finite partial sums. Indeed, $\|\iota_{\ell_1(I)}(e_i)^+\| \leq \|\iota_{\ell_1(I)}(e_i)\| = 1$, so the net over finite subsets of I is absolutely Cauchy and

$$\|Sa\| \leq \sum_{i \in I} |a_i| = \|a\|_{\ell_1(I)}.$$

Thus $S : \ell_1(I) \rightarrow \text{FBL}[\ell_1(I)]$ is a contraction. It is positive because nonnegative coordinates are sent to a norm-convergent sum of positive free-lattice elements.

Since ρ is a lattice homomorphism,

$$\rho(\iota_{\ell_1(I)}(e_i)^+) = \rho(\iota_{\ell_1(I)}(e_i))^+ = e_i^+ = e_i.$$

By linearity and continuity, $\rho Sa = a$ for every $a \in \ell_1(I)$. Hence $\ell_1(I)$ positively factors through $\text{FBL}[\ell_1(I)]$ with $\|S\| \|\rho\| \leq 1$. $\square$

Theorem 1. *Let X be a Banach lattice. Suppose that X is a positive retract of $\ell_1(I)$: there are positive bounded linear maps*

$$J : X \rightarrow \ell_1(I), \quad Q : \ell_1(I) \rightarrow X, \quad QJ = I_X.$$

Then X positively factors through $\text{FBL}[X]$. More precisely, there are positive maps

$$U : X \rightarrow \text{FBL}[X], \quad V : \text{FBL}[X] \rightarrow X,$$

such that $VU = I_X$ and

$$\|U\| \|V\| \leq \|J\| \|Q\|.$$

Proof. Let $S : \ell_1(I) \rightarrow \text{FBL}[\ell_1(I)]$ be the positive contraction from Lemma 1. Let

$$\text{FBL}[Q] : \text{FBL}[\ell_1(I)] \rightarrow \text{FBL}[X]$$

be the lattice homomorphism induced by Q , and let

$$V = \rho_X : \text{FBL}[X] \rightarrow X.$$

Define

$$U = \text{FBL}[Q] S J : X \rightarrow \text{FBL}[X].$$

The map U is positive because J and S are positive and $\text{FBL}[Q]$ is a lattice homomorphism. Also

$$\|U\| \leq \|\text{FBL}[Q]\| \|S\| \|J\| \leq \|Q\| \|J\|, \quad \|V\| \leq 1.$$

It remains to check that $VU = I_X$. Let $x \in X$, and put $a = Jx = (a_i)_{i \in I} \in \ell_1(I)$. Using the norm-convergent expansion in Lemma 1,

$$VUx = \rho_X \text{FBL}[Q] \left(\sum_{i \in I} a_i \iota_{\ell_1(I)}(e_i)^+ \right).$$

Since $\text{FBL}[Q]$ and ρ_X are lattice homomorphisms,

$$\rho_X \text{FBL}[Q] (\iota_{\ell_1(I)}(e_i)^+) = \rho_X (\iota_X(Qe_i)^+) = Qe_i,$$

where the last equality uses $Qe_i \geq 0$, because Q is positive. Therefore

$$VUx = \sum_{i \in I} a_i Qe_i = Q(Jx) = x.$$

So X has the claimed positive factorization through $\text{FBL}[X]$. □

Corollary 1. *Every finite-dimensional Banach lattice positively factors through its free Banach lattice.*

Proof. A finite-dimensional Banach lattice is lattice isomorphic to $\mathbb{R}^n$ with coordinatewise order. Let J be the positive coordinate map into ℓ_1^n , and let Q reconstruct the corresponding linear combination of the positive atoms in X . Then J and Q are positive bounded maps and $QJ = I_X$. Theorem 1 applies. □

Verification Notes

The only infinite-sum point is in Lemma 1. It is controlled by

$$\sum_i |a_i| \|\iota_{\ell_1(I)}(e_i)^+\| \leq \sum_i |a_i|,$$

so the partial sums converge in norm even for arbitrary index sets. The identity $VU = I_X$ is checked first on this absolutely convergent expansion and then follows by continuity. No numerical computation is used.

The result also combines with Corollary 3.6 of the source paper: if such an X has the μ -approximation property, then it has the $\mu \|J\| \|Q\|$ -positive approximation property. That consequence is not the main claim here; the packet records the positive factorization itself.

Limitations And Novelty Check

This packet does not solve the source paper’s general question. The proof requires a positive retraction through an atomic $\ell_1(I)$. It gives no answer for arbitrary Banach lattices, nonatomic L_1 -spaces, c_0 , or Banach lattices where no such atomic positive lifting is known.

Before packaging, the run’s lightweight indexes were searched for arXiv:2303.05209, the title, positive factorization, AP/PAP, and free Banach lattice phrases. No exact duplicate packet or prior attempt was found. A bounded web search on June 14, 2026 for exact phrases around positive factorization through $\text{FBL}[X]$ did not locate a later paper explicitly answering the question. This should be treated as a candidate partial result for human review, not as a full literature-status claim.

Human Review Recommendation

The proof is short and appears likely valid. The main points to check are the definition of S on $\ell_1(I)$, the use of the positive part of each free generator, and the transfer step using positivity of Q :

$$\rho_X \text{FBL}[Q](\iota(e_i)^+) = Qe_i.$$

References

- [1] T. Oikhberg, *Geometry of unit balls of free Banach lattices, and its applications*, arXiv:2303.05209, 2023.
- [2] A. Avilés, J. Rodríguez, and P. Tradacete, *The free Banach lattice generated by a Banach space*, Journal of Functional Analysis 274 (2018), 2955–2977.
- [3] A. Avilés, G. Martínez-Cervantes, J. Rodríguez, and P. Tradacete, *A Godefroy-Kalton principle for free Banach lattices*, Israel Journal of Mathematics 247 (2022), 433–458.